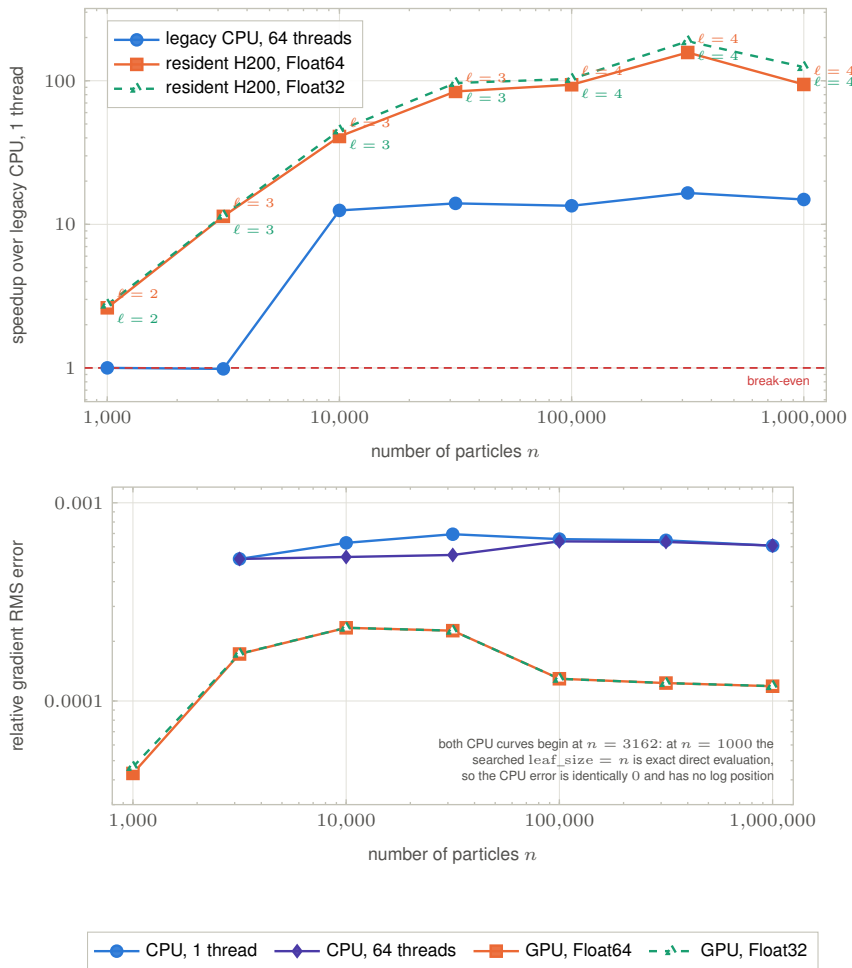


Fig. 9 — End-user CPU and GPU scaling at literature $P = 4$



Regime: $n \in \{1000, 3162, 10^4, 31623, 10^5, 316228, 10^6\}$, literature $P = 4$ means code `expansion_order=3` (degrees 0:3). The legacy octree fixes $\theta = 0.5$ and independently searches leaf size for each $(n, \text{threads})$ on a step-5 refinement grid; its timed call includes tree and interaction-list construction. It also disables threading outright below $n = 5000$ (`src/fmm.jl` forces one thread when $n_{\text{target}} + n_{\text{source}} < \text{MIN_BODIES} = 10^4$), which is why the 64-thread curve sits at $1\times$ for $n \leq 3162$ and jumps at $n = 10^4$. The resident H200 path excludes one-shot cache construction, prefers dense translation, and selects the fastest measured ℓ per point. Its reviewed analytic stencil uses $\epsilon_\ell = 0.195423853310349172^{\ell-4}$, matching the legacy strict equal-cell cutoff (squared offset 12 rejected, 13 accepted); adaptive interaction lists are not structurally identical. At $n \geq 10^5$, the planned $\ell = 6/7$ caches exceeded H200 capacity even after the prescribed precomputed-Y fallback; those eight attempts are retained in the failure ledger, and selection uses the 36 successfully constructed cases. Float64 GPU errors must be within $10\times$ of both CPU errors whenever the CPU result uses an FMM decomposition (`leaf_size < n`); exact/direct CPU cases are reported but exempt from that ratio gate. Float32 ratios are recorded but not gated. Curves use minimum post-warmup step time; the lower panel uses the shared sampled-direct references. **Data:** task 024b, `data/cpu_gpu_scaling/*.csv`.